Ava E. Bush, PhD

Overview

Earth & geospatial scientist with over a decade of research experience. Passionate about bridging science and practical application while advancing community-centered climate education.

Contact

entradaava@gmail.com | 925-342-0725

Education

Ph.D. Earth System Science | Stanford University | 2024

M.S. Marine Science | University of San Diego | 2016

B.S. Earth & Environmental Science | Furman University | 2013

Research & Professional Experience

Climate Change Impacts | Stanford University | June 2018 - April 2024
Stanford University provided funding

- Investigated impacts of climate variability on Earth surface processes
- Designed reproducible workflows for geospatial modeling, statistical analysis, machine learning, and data visualization
- Applied principles of econometrics to infer causality
- Analyzed remotely-sensed data products to detect patterns and trends
- Published doctoral dissertation and peer-reviewed journal article

Isotope Geochemistry | Stanford University | August 2017 - June 2018

- Researched isotopic signatures of tectonically-driven landscape evolution

Trace Metal Geochemistry | University of San Diego | Aug. 2014 - May 2017
Dean's Merit, Stephen Sullivan Memorial, and San Diego Association of Environmental Professionals Scholarships provided funding

- Studied ancient volcanic system trace metal geochemistry
- Processed and visualized geospatial and geochemical data
- Mentored high school, undergraduate, and graduate students
- Published master's thesis; presented at international conferences

Physical Science Lab Assistant | Univ. of San Diego | May 2015 - Aug. 2016
Employed by Environmental and Ocean Sciences Department

- Ensured proper safety protocol and data collection, including quality assurance/quality control (i.e., QA/QC) for physical science courses (e.g., Environmental and Ocean Geochemistry)
- Resolved technical concerns with laboratory instrumentation

Energy Policy Analyst | University of San Diego | December 2013 - May 2015
Employed by Energy Policy Initiatives Center

- Developed greenhouse gas inventories to estimate emissions among the electricity, transportation, waste, water, and agricultural sectors
- Co-authored climate action plans that quantified potential emissions reductions and made recommendations for climate policies

Climate Education | University of San Diego | October 2013 - March 2014
Employed by Energy Policy Initiatives Center

- Synthesized climate effects on extreme events, water, and public health
- Conducted standardized interviews about knowledge of local climate impacts with community leaders among public and civil society sectors
- Coordinated with local nonprofits (e.g., SANDAG, San Diego Foundation)

Engineering Communications | Vanderbilt University | May - August 2013
Employed by Vanderbilt University Graduate School of Engineering
- Wrote articles and created informational material for School of Engineering

Geospatial Modeling | Furman University | Aug. 2012 - May 2013
Furman Advantage Fellowship provided funding
- Applied ArcGIS and remotely sensed imagery to assess landslide hazard
- Presented at international conference and was covered in local news

Carbon Mitigation | Upstate Forever | May - August 2011
Shi Sustainability Fellowship provided funding
- Analyzed and reported the feasibility of a local carbon offsets program
- Conducted literature reviews, held focus groups with local businesses, and distributed an online survey which received over 500 responses

Fluvial Sediment Transport | Furman University | August 2009 - May 2010
Howard Hughes Medical Institute (HHMI) provided funding
- Researched particulate organic carbon transport via first-order streams

Teaching Experience

Academic Skills Educator | Stanford University | June - October 2022
- Co-developed and led workshops that centered planning and resilience

Teaching Assistant | Stanford University | September 2017
- Field assistant for Geological History of the Rocky Mountains

Instructor | University of San Diego | August 2015 - May 2017
- Taught lab sections for Natural Hazards (4 semesters) and Environmental Geology (1 semester)

Consultant | Furman Center for Teaching & Learning | Aug. 2010 - May 2013
- Provided writing and multimedia support to undergraduate students

Leadership, Service, Honors & Awards

2023 | **PhD Pathways Advisory Committee** | Stanford Career Education
2022 | **Mentor** | Girls in STEM
2021 | **Stanford Earth Outstanding Service Award for DEI** | Stanford University
2019 | **Committee** | Women in Data Science @ Stanford Earth Conference
2018 – 2019 | **Executive Board** | Stanford University Women in Earth Science
2015 | **San Diego Association of Environmental Professionals** Scholar
2015 | **University of San Diego** | Dean's Merit Scholar
2015 | **National Science Foundation** | GRFP Honorable Mention
2013 | **Furman University** | *magna cum laude*

Computational Skills

Programming Languages: statistics/data-science (R, Python); numerical computing/engineering (MATLAB); compiled (C); scripting (bash)

Geospatial Analysis: ArcGIS, QGIS, Google Earth Engine

Version Control & Repositories: Git/Github, Jupyter Notebooks, Zenodo

Data Management & Visualization: Adobe Creative Suite; Microsoft Office Suite (e.g., Excel, PowerPoint); Google Suite (e.g., Sheets, Slides)

Bush, Ava E. (formerly Elizabeth C. Johnston)

Dissertations & Theses

- **Johnston, E.** (2024). *Impact of climate variability on landslide hazard in the western United States*. Doctoral Dissertation, Stanford University.
- **Johnston, E.** (2016). *Geochemical investigations of [trace metal] enrichment in the Santiago Peak Volcanics*. Master's Thesis, University of San Diego.

Peer-Reviewed Publications

- **Johnston, E.**, Davenport, F., Willenbring, J., Burke, M., & Diffenbaugh, N. (2021). Quantifying the effect of precipitation on landslide hazard in urbanized and non-urbanized areas. *Geophysical Research Letters*, 48.

Conference Presentations and Abstracts

- **Johnston, E.**, Davenport, F., Willenbring, J., Burke, M., & Diffenbaugh, N. (2021). Effect of wildfire severity on the longevity of precipitation-triggered landslide hazard. AGU Fall Meeting (Poster).
- **Johnston, E.**, Burke, M., Caers, J., Davenport, F., Wang, L., Muthukrishnan, S., & Diffenbaugh, N. (2020). Impact of urbanization on the relationship between antecedent precipitation and landslides: Evidence from the San Francisco Bay Area. AGU Fall Meeting (Talk).
- **Johnston, E.**, Burke, M., Caers, J., Davenport, F., Wang, L., Muthukrishnan, S., & Diffenbaugh, N. (2019). Combined influence of antecedent rainfall and urbanization on landslides: Empirical evidence from the U.S. Pacific Coast. AGU Fall Meeting (Talk).
- **Johnston, E.**, Caers, J., Wang, L., Davenport, F., Muthukrishnan, S., & Diffenbaugh, N. (2018). Multi-scale signatures of climate change on landslide susceptibility along the U.S. Pacific Coast. AGU Fall Meeting (Poster).
- **Johnston, E.**, Pollock, M., Cathcart, E., AlBashaireh, A., & O'Shea, B. (2016). Geochemical controls on anomalous arsenic enrichment in the Santiago Peak Volcanics, Southern California. AGU Fall Meeting (Poster).
- AlBashaireh, A., **Johnston, E.**, O'Shea, B., & Pollock, M. (2016). Arsenic extent and mobilization in soil and vegetation near abandoned ultra-enriched arsenic mines, Black Mountain Open Space Park, San Diego. GSA Annual Meeting (Poster).
- **Johnston, E.**, Cathcart, E., Robinson, T., Phillips, C., & O'Shea, B. (2015). Arsenic enrichment associated with historic gold mining in Julian, CA. GSA Cordilleran Section Meeting (Poster).
- **Johnston, E.**, & Silva-Send, N. (2015). Urban climate action planning: Greenhouse gas mitigation tool for local energy and climate policy analysis. Climate Change Impacts and Responses (Talk).
- O'Shea, B., Robinson, T., **Johnston, E.**, & Cathcart, E. (2014). Geogenic arsenic in metamorphic bedrock and impacts from historic gold mining near Julian, CA. GSA Annual Meeting (Talk).
- Ranson, W., Muthukrishnan, S., **Johnston, E.**, Campbell, S., & Geier, P. (2013). Ongoing research in slope stability in Upstate South Carolina. GSA Annual Meeting (Poster).
- **Johnston, E.**, & Muthukrishnan, S. (2013). Landslide hazard zonation using GIS-based modeling. GSA Southeastern Section Meeting (Poster).
- **Johnston, E.**, & Anderson, B. (2011). Particulate organic carbon transport in headwater streams of the South Carolina Piedmont. GSA Southeastern Section Meeting (Poster).